SHUBHAM SHARMA

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Research Interests

My goal is to develop robots with human-like intelligence which includes embodying them with congnitive capabilities for understanding reality and physical interlligence to influence it. Specifically, I am interested in:

- Learning robust sensorimotor skills (dexterous manipulation, whole-body control, visuotactile manipulation)
- Adapting learned skills to new environments (rapid adaptation, continual learning)
- Composing skills for more complex tasks (skill-chaining, energy-based models)

Education

Indian Institute of Technology Bombay — Mumbai, India

Bachelor of Technology in Aerospace Engineering (with Honors) — *July 2019 – Aug 2023*

- Cumulative GPA: **8.68/10.0**
- Minor Degree: Artificial Intelligence and Data Science

Publications

1. *Haoyi Niu,**Shubham Sharma**,Yiwen Qiu,Ming Li,Guyue Zhou,Jianming Hu,Xianyuan Zhan.* "When to Trust Your Simulator: Dynamics-Aware Hybrid Offline-and-Online Reinforcement Learning". **NeurIPS 2022** (**Spotlight**) [arXiv] [Code].

2. *M.Agarwal,R.Shukla,K.Badoriya,R.Khantwal,H.Naik,M.Dixit,A.Mete,**S.Sharma**,A.Sonara,B.Prakash, D.Shukla.*A Novel Approach to Autonomous Aerial Manipulation using Multiple Drones". *IARC 2021 Symposium on Autonomous aerial flight*

Research Experience

Robot learning | Cognitive Learning for Vision and Robotics Lab — *July 2023 - Present*

Guide: Prof. Joseph Lim, Kim Jaechul School of Artificial Intelligence — *KAIST*

- Primarily worked on building scalable long-horizon memory systems for decision making agents; created new sim and real world tasks and implemented SOTA baselines for the same
- Also worked on motion planning solutions for automating precision welding on a real robot

Offline Reinforcement learning | DREAM Lab — *Jan 2022 - Dec 2022*

Guide: Prof. Xianyuan Zhan, Institute for AI Industry Research (AIR) — *Tsinghua University*

- **Introduction:** *Learning effective reinforcement learning (RL) policies for the real world is tough without a high-fidelity simulator or huge amounts of collected data. Can we combine learning from limited real world data in offline RL and unrestricted exploration in imperfect simulators to address the problems of real world learning?*
- Developed **H2O**, a hybrid Offline-Online RL algorithm which adaptively penalizes the Q function update on simulated state-action pairs with large dynamics gaps, simultaneously learning from a fixed real-world dataset.
- Empirical evaluations in simulated and real-world environments, coupled with theoretical analysis, demonstrate H2O's superior performance over existing cross-domain online and offline RL algorithms.

Open World Semi-Supervised Learning | Bachelor's Thesis — *Oct 2022 - Apr2023*

Guide: Prof Biplab Banerjee, Center for Machine Learning nad Data Science — *IIT Bombay*

- **Introduction:** *the real world is inherently dynamic and open — new classes can emerge in the test data that have never been encountered during training. Can we build models that can either classify instances into one of the previously seen classes or discover new classes and assign instances to them?*
- Studied the effects of noisy Pseudo Labels in discovering new classes from the unlabelled data.
- Developed a novel method involving student-teacher distillation for Open World SSL

Multi agent collision avoidance | Supervised Project [Final Report] — *Jan 2021 - Jul 2021*

Guide: Prof Leena Vacchani, Systems and Controls Engineering — *IIT Bombay*

- **Introduction:** *Collision detection & avoidance are vital sub-tasks in any autonomous agent. Can we create low cost algorithms on FPGAs and handle multiple agents to ensure collision avoidance in dynamic environments?*
- Worked on the simulating **obstacle avoidance** for a system of mobile robots using **Interval Analysis** techniques
- Formulated a new mathematical model for the intervals approach; concept proof tested using ROS in GAZEBO
- Studied the hardware expandability of the approach for parallel and embedded applications in robotic systems

Professional Experience

Deep Learning Intern | Daikin, Japan | [Demo] *May 2022 – July 2022*

Worked with the RnD team to develop models for ***multilingual document understanding***

- Created the **training pipeline** for the model; used the **transformer** based **LayoutXLM** as the backbone model
- Achieved **zero-shot transfer** to Japanese forms via a **question answering system** trained on multilingual data
- Improved zero-shot accuracy of **74% to 95%** by **few-shot training** on Japanese language using only **9** samples

RnD Intern | Space x View Pte. Ltd, Singapore *May 2020 – Sep 2020*

The firm works on developing autonomous systems and services for the east Asian Space industry

- Worked with a team of **5** co-interns to create the **first working prototype** of a system of controllable buoys capable of oceanic data collection for building real-time models that help the **Japanese Fishing Industry**
- Tasked with the design of the **Power Management System** for the buoy to make it operational for **3+ hrs**
- Integrated the **PH, Temperature, TDS and GPS** sensors along with the **Solar Panels** on the final prototype

Technical Leadership

Team Leader | Project AeRoVe | Innovation Cell IIT Bombay *Sep 2019 - Present*

Introduction: *Leading the technical forefront of a multidisciplinary team of* ***20*** *students working on developing autonomous quadcopters for Aerial Robotics competitions all around the globe*

Research	• **Motion Planning**: Implemented algorithms like **A*, RRT* & DWA** for **indoor drone systems** • **Controls**: Developed real-time **minimum snap trajectories** for the drone using **MAVROS** scripts • **Computer Vision**: Optimised **YOLOv4** using **TensorRT** engines to speed up inference by **300%** • **Localization**: Deployed **ORB-SLAM2** and **GPS-RTK** for real time pose estimation in drones
Impact	• Declared **WORLD CHAMPIONS** for IARC Mission 9 Simulation Challenge, hosted by **AUVSI** • Acheived **WORLD RANK 4** in the **Student Design Challenge 2021**, hosted by **ASME** • Secured **third place** in **DRDO**'s navigation challenge among 12 IITs in the **InterIIT Tech 2021**

Other Technical Projects

Muscle Memory: retrieval augmented RL for Dexterous Manipulation | [Report] *Fall 2023*

- Extended retrieval RL to dexterous manipulation tasks, improving RL sample efficiency in high-dim control.
- Developed a two-phase approach: (1) Play phase - extracting action abstractions from multi-task data to create a lookup table, (2) Online learning phase - trained RL agents using standard setup augmented with the retrieved action abstractions.

GPU acclerated Multi-Agent Trajectory Optimization | *CS769: Optimization for ML* | **[Code]** *Spring 2022*

- Worked on the problem of generating collision free trajectories for multiple agents under collision constrains.
- Reformulated the classical setup as a simple convex case and solved the **quadratic programming** problem
- Parallelized the whole piepline using **JAX** and reduced the inference time in the original implementation by **57.1%**

RNN Repair | *IE635: Deep Learning* | **[Final Report]** *Fall 2021*

- Worked on repairing RNN misclassifications by using a light-weight model-based **influence analysis** technique
- Performed **Sample-level** and **Segment-level** influence analysis of RNNs tested on the **Toxic** dataset
- Extended the Sample-level Influence Analysis to **DNNs** and proof tested the hypothesis on the **MNIST** dataset

Extended Mask R-CNN | *GNR638: Machine Learning for remote sensing* | **[Code]** *Fall 2021*

- Constructed a two step approach for **multi-label classification** of **7** different Yoga Poses on a custom dataset

- Used the Mask R-CNN trained on the **COCO** dataset as the backbone model for human **instance segmentation**
- Developed a **neural network** for classifying the extracted instances to achieve a test accuracy of **95.6%**

Reinforcement Learning | *CS747: Foundations of Intelligent and Learning Agents* *Fall 2021*

- Implemented **ϵ-greedy, UCB, KL-UCB** and **Thompson Sampling** algos to stochastic Multi-armed bandits
- Used **Value iteration** and **Policy iteration** to compute an optimal policy for an agent playing Anti-Tic-Tac-Toe
- Solved the **Mountain Car problem** using SARSA with Linear Approximation using **Tile Coding**

ASME - Student Design challenge 2021 | *Innovation Cell* *Spring 2021*

- Reached the **World Finals** of the competition after going through several knockout rounds | GLOBAL RANK 4
- Designed and created a stack of **Boost Converters** on **customized PCBs** to maximise battery performance
- Constructed a custom **differential drive** for the robot using SPDT Relays, ATTINY IC & Bluetooth module

International Aerial Robotics Challenge 2021 | *Innovation Cell* *Spring 2021*

- Implemented dynamic **Waypoint Navigation** using **ROS** and **GAZEBO** for a mother-daughter drone system
- Developed an algorithm to **localize** points in the UAV environment using only the feed of a **monocular camera**
- Formulated and implemented a **Finite-State-Machine** structure acting as the central **Autopilot** for the drones

Project ETH | *Innovation Cell* | **[Code]** *Fall 2020*

- Created a real-time simulation of a robot performing a series of Automated tasks to solve the problem statement
- Extracted and manipulated the **Point Cloud** data to localize the interest points using **TF Trees** in **ROS**
- Formulated **Inverse Kinematics** on a 3-Degree Of Freedom **robotic arm** system to be used by the robot

Technical Skills

Programming Languages: C++, Python, MATLAB, Bash
Machine Learning: PyTorch, TensorFlow, Keras, OpenCV, Numpy, Pandas, Seaborn
Software: Docker, LaTeX, Git, AutoCAD, SolidWorks, Arduino IDE

Key Coursework

Computer Science: Data Structures and Algorithms, Machine Learning for Remote Sensing, Deep Learning, Foundations of Intelligent and Learning Agents, Optimization for Machine Learning, Deep Learning for NLP

Robotics: Control Theory, Guidence and Navigation, Motion Planning & coordination of Autonomous Vehicles

Mathematics: Differential Equations, Calculus, Linear Algebra, Numerical Analysis

MOOCs: CS229-Machine Learning (Stanford), CS330-Deep Multi-Task & Meta learning, CS285-Deep RL (Berkeley)

Mentoring Roles

Academic Mentor | Student Mentorship Program, IIT Bomaby *Spring 2021 - Present*

- Part of a **23** member team of mentors selected based on interviews & peer-reviews, mentoring **7 sophomores**
- Helping mentees, strike a balance between **academics & extracurriculars**, and manage time efficiently
- Contributed in the **Department Video Initiative**, aimed as an introduction to the department labs & facilities

Extra Curricular Activities

Social	• Mentored **25+** middle school students towards building a solar lamp with **SoULS** IIT Bombay • Created 2+ hrs of educational **YouTube** content for underprivileged children with **NSS**NSS
Sports	• Received 1 yr of professional training in **Football** under the **National Sports Organization** • received **Bronze** in the Dehradun district during the **Ranger's U-17 Cup** with the **BBA**
Cultural	• Stood **First** among all the freshmen in the Inter Hostel **Music Video Making Competition** • Performed a group **hip-hop** dance in front of a crowd of **500+** people at **Salsa Night**